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Ex 9: Travelling Salesman Problem (TSP)

CODE:

```
from itertools import permutations


def tsp(distance_matrix):
    n = len(distance_matrix)
    cities = list(range(1, n))
    min_cost = float('inf')
    best_path = []

    for perm in permutations(cities):
        cost = 0
        current = 0
        for city in perm:
            cost += distance_matrix[current][city]
            current = city
        cost += distance_matrix[current][0]

        if cost < min_cost:
            min_cost = cost
            best_path = [0] + list(perm) + [0]

    return min_cost, best_path
```

```
matrix = [
   [0, 10, 15, 20],
   [10, 0, 35, 25],
   [15, 35, 0, 30],
   [20, 25, 30, 0]
]
cost, path = tsp(matrix)
print("Minimum Cost:", cost)
print("Optimal Tour:", path)
```

OUTPUT:

```
============== RESTART: C:/Users/Saiko/Documents/CSA1738/EXP-9.py =========
Minimum Cost: 80
Optimal Tour: [0, 1, 3, 2, 0]
```